



Forecasting Admin Page — Technical Documentation

Coresight Research Portal — Technical Documentation

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Application: Market Data Portal (MDP)

Source: `app/pages/forecasting_admin.py`

Documented URL: `/forecasting_admin` (per `CLAUDE.md`)

Purpose: Bulk sync of stored revenue model forecasts into `coreiq_model_forecasts` when annual actuals are newer than cached forecasts.

Registration Status

Important: `forecasting_admin.py` exists under `app/pages/` but is **not listed** in `app/main.py` `_PAGES_CONFIG` as of current source. The page is therefore **not registered in Streamlit navigation** unless added manually or loaded via an alternate entry point.

Related surfaces:

- `app/pages/admin_iam.py` — unregistered duplicate Identity and Access Management (IAM) patterns
- Embedded Identity and Access Management (IAM) expander **inside** `forecasting_admin.py` itself
- `access_management.py` — canonical Identity and Access Management (IAM) for `forecasting_admin` page Access Control List (ACL)

Operators may need to add:

```
("pages/forecasting_admin.py", "Forecasting Admin", "forecasting_admin", False),
```

to `_PAGES_CONFIG` for `/forecasting_admin` to resolve.

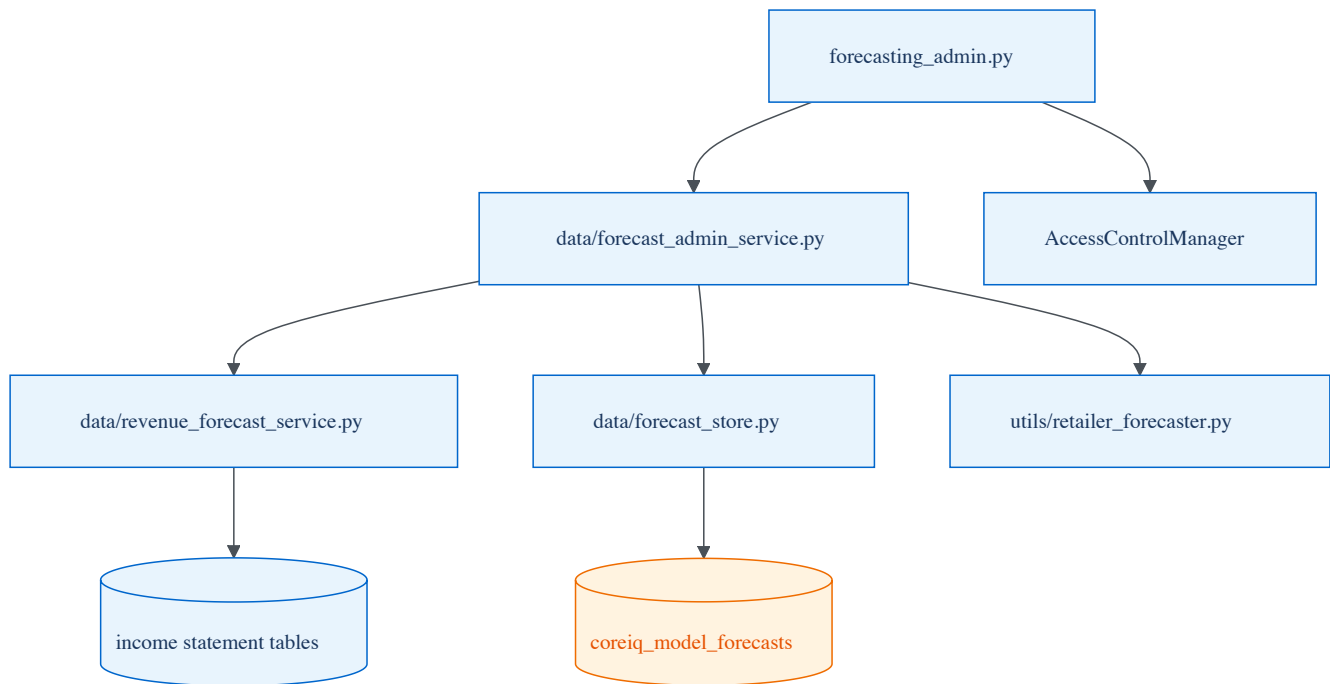
Overview

Forecasting Admin is an **internal write path** for the revenue forecasting pipeline (annual and quarterly). While `forecasting.py` is read-only for most users, this page triggers:

1. **Single-ticker sync** — `sync_forecast_for_ticker` (annual) or `sync_quarterly_forecast_for_ticker`
2. **Bulk stale sync** — `sync_all_eligible` or `sync_all_eligible_quarterly`
3. **Optional embedded IAM** — manage ACL for admin pages (hardcoded admin emails only)

After sync, `clear_revenue_forecast_caches()` invalidates Streamlit caches so Market Data / Estimates pick up new rows.

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Authentication & Access Control

Auth

```
user_email = get_current_user()
```

Identity is established via `main.py` cookie hydrate. Page-level `require_auth()` may be enabled for direct-route protection.

Sync permission

```
has_access = AccessControlManager.check_access("forecasting_admin", user_email, "allow")
```

Cached in session:

```
_cache_key = f"fc_admin_access_{user_email.lower()}"
st.session_state[_cache_key] = has_access
```

Requires permission level $\geq$ `allow` (rank 4) **or** `admin` / `super_user` role bypass.

Users without access see warning: sync buttons hidden; IAM expander may still show for hardcoded admins.

IAM expander admins

Separate from DB roles — hardcoded set:

Email
mohdsaeedafri@coresight.com
shashankgupta@coresight.com
risthavilinadsa@coresight.com
philipmoore@coresight.com

When `is_admin` (email in set), expander **Access Control — Page Permissions** is visible.

Legacy allowlist

`forecast_admin_service.FORECAST_ADMIN_EMAILS` (4 emails) + `is_forecast_admin()` — **not used** by current `forecasting_admin.py` UI (ACL replaced it).

UI Layout

Chrome

- `new_rerun_id("forecasting_admin")`
- `st.set_page_config(page_title="Forecasting Admin", layout="wide")`
- `hide_sidebar()`, `render_styles()`, `inject_red_spinner_css()`
- `render_header(current_page="forecasting_admin")`
- Custom CSS: `.fc-admin-hero`, cards, deny state, bold form labels

Sections

1. **Hero** — "Internal · Forecast Store" / "Forecasting Admin"
2. **IAM expander** (admin emails only) — mini access management
3. **Sync controls** (if `has_access`)
4. **Results table + metrics** (after sync run)
5. **Page load diagnostics** expander — timing breakdown
6. `render_coresight_footer`

Embedded IAM Expander

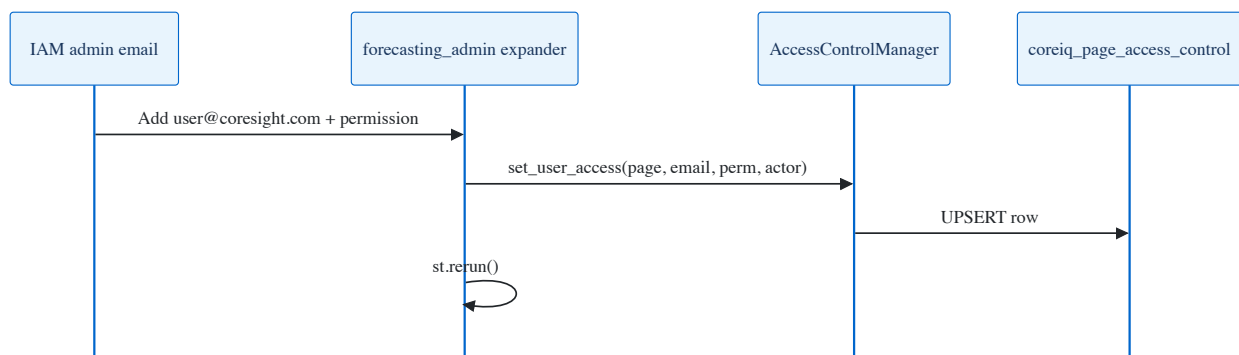
Duplicates patterns from `access_management.py` for three pages:

```
MANAGED_PAGES = ["forecasting_admin", "retailer_adding", "company_filings_add_files"]
```

Features:

- Page selector → `get_page_users(selected_page)`
- Stats: Total Users, Allow Access, Delete Admin
- Add user form → `set_user_access`
- Per-user permission select + Save / Delete
- Permission legend (allow, deny, view_only, upload_only, edit, delete_admin)

- *Permission legend (allow, deny, viewonly, uploadonly, edit, deleteadmin)*



Sync Operations

Sync all stale tickers

```
run_all = st.button("Sync all stale tickers")

results = sync_all_eligible(force=False, periods=5, progress_callback=_cb)

clear_revenue_forecast_caches()
```

- Progress bar via callback `(cur, total, display_ticker)`
- `force=False` — respects `needs_update()` staleness check

Sync single ticker

```
run_one = st.button("Sync this ticker only")

row = sync_forecast_for_ticker(ticker.strip())

clear_revenue_forecast_caches()
```

Enriches result with `company_name` from `RevenueForecastService.get_companies()`.

Result metrics

Metric	Counts status
Upserted	<code>updated</code>
Already current	<code>up_to_date</code>
Skipped / no data	<code>skipped</code> , <code>no_data</code>
Errors	<code>error</code>

Dataframe columns: `ticker`, `company_name`, `status`, `rows_upserted`, `message`

Service Layer — `forecast_admin_service.py`

```
sync_forecast_for_ticker(display_ticker, force=False, periods=5)
```

Diagram — Part A: Source gate & actuals prep

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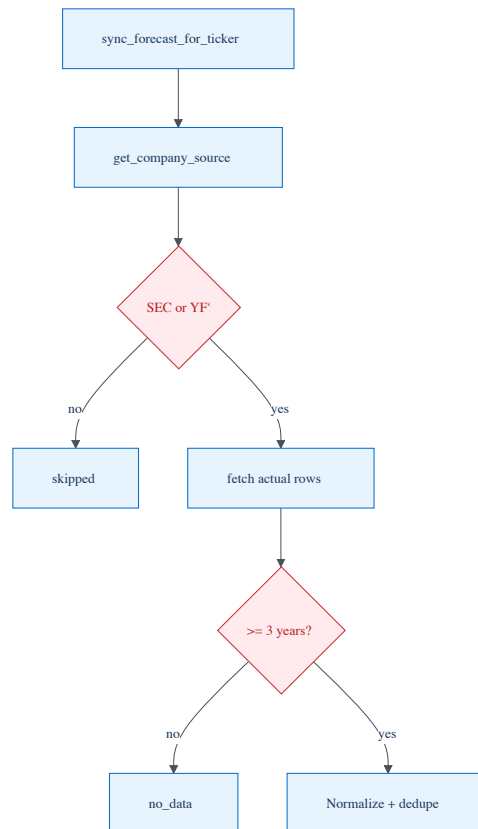
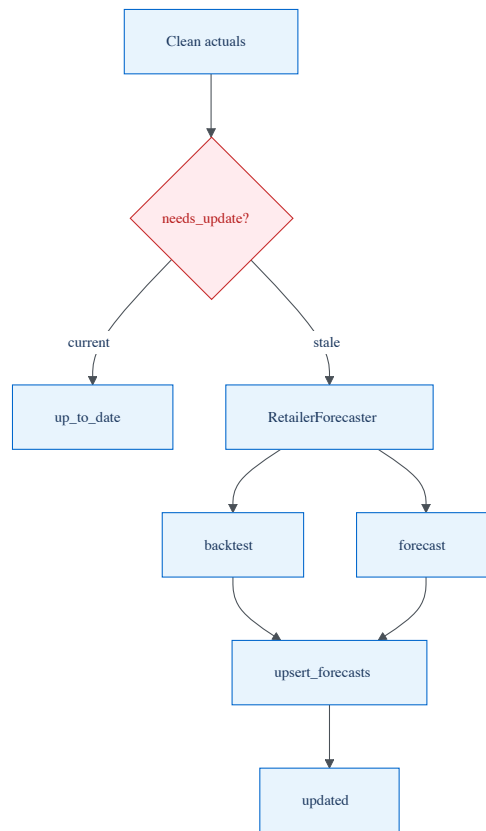


Diagram — Part B: Staleness check & forecast store

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Return dict keys: `ticker`, `status`, `message`, `rows_upserted`, `store_ticker`, `last_actual_date`, `company_name`, `exchange`

Status values:

Status	Meaning
<code>updated</code>	Engine ran; DB upserted
<code>up_to_date</code>	<code>needs_update</code> false and not forced
<code>skipped</code>	Unsupported source or insufficient years
<code>no_data</code>	No revenue rows
<code>error</code>	Exception captured

```
sync_all_eligible(force, periods, progress_callback, exclude_tickers)
```

Iterates `RevenueForecastService.get_companies()` — same universe as Revenue Estimates (≥ 3 annual revenue points).

```
clear_revenue_forecast_caches()
```

```
RevenueForecastService.get_company_dashboard.clear()
RevenueForecastService.get_companies.clear()
```

Database — `coreiq_model_forecasts`

Written by `forecast_store.upsert_forecasts()`.

Staleness — `needs_update(ticker, last_actual_date)`

Compares latest annual actual `fiscal_date_ending` against `MAX(last_actual_date)` stored per ticker/metric.

Also forces recompute if `computed_at < 2026-05-11` (dedup fix cutoff `_DEDUP_FIX_DATE`).

Key columns (conceptual)

Column	Purpose
<code>ticker</code>	Composite for YFinance (e.g. <code>TSCO.L</code>)
<code>fiscal_year</code>	Forecast year
<code>model_key</code>	linear, cagr, ensemble, scenario_*, etc.
<code>value_millions</code>	Forecast value
<code>metric</code>	<code>total_revenue</code>
<code>mape</code>	Backtest accuracy
<code>last_actual_date</code>	Actuals snapshot used
<code>computed_at</code>	When forecast was stored
<code>company_name</code> , <code>exchange</code>	Denormalized (via <code>forecast_refresh_service</code>)

Source actuals tables

Source	Table
SEC	<code>coreiq_av_financials_income_statement</code>
YFinance	<code>coreiq_yf_financials_income_statement</code>

Forecasting Engine

`utils.retailer_forecaster.RetailerForecaster` :

- Six models: linear, cagr, exp_smoothing, holt, ma_trend, weighted_avg
- `backtest(holdout_years=2)` — ranks by MAPE
- `forecast(periods)` — includes ensemble (top 3 MAPE) + scenario percentiles
- Data cleaning: duplicate year dedup, 10% of max revenue floor, outlier detection

Session State

Key	Purpose
<code>fc_admin_access_{email}</code>	Cached ACL boolean
<code>iam_page_select</code> , <code>iam_new_email</code> , <code>iam_new_perm</code>	IAM expander widgets
<code>iam_perm_{page}_{email}</code>	Per-user permission selectbox state
<code>fc_one_ticker</code>	Single ticker input
Widget keys for sync buttons	<code>fc_sync_all</code> , <code>fc_sync_one</code>

Performance Instrumentation

Page records `_timings` list:

- `hide_sidebar + render_styles`
- `get_current_user`
- `render_header`
- `check_access`
- `render_footer`

Displayed in expander: **Page load diagnostics — total Nms** with color-coded bars (>300ms red).

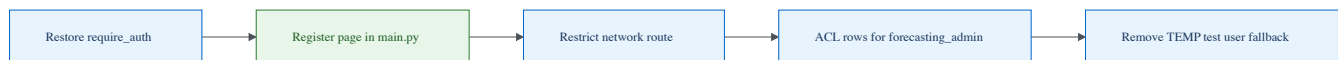
Relationship to `forecasting.py`

Aspect	<code>forecasting.py</code>	<code>forecasting_admin.py</code>
Primary mode	Read / visualize	Write / sync store
DB writes	Via refresh dialog (edit ACL)	Bulk admin sync
ACL page name	<code>forecasting</code>	<code>forecasting_admin</code>
Registered in main.py	Yes	No (currently)
Cache clear after write	Yes (<code>clear_revenue_forecast_caches</code>)	Yes

The Estimates page **Refresh Data** dialog calls the same `sync_forecast_for_ticker` / `sync_all_eligible` for users with `edit` on page `forecasting` .

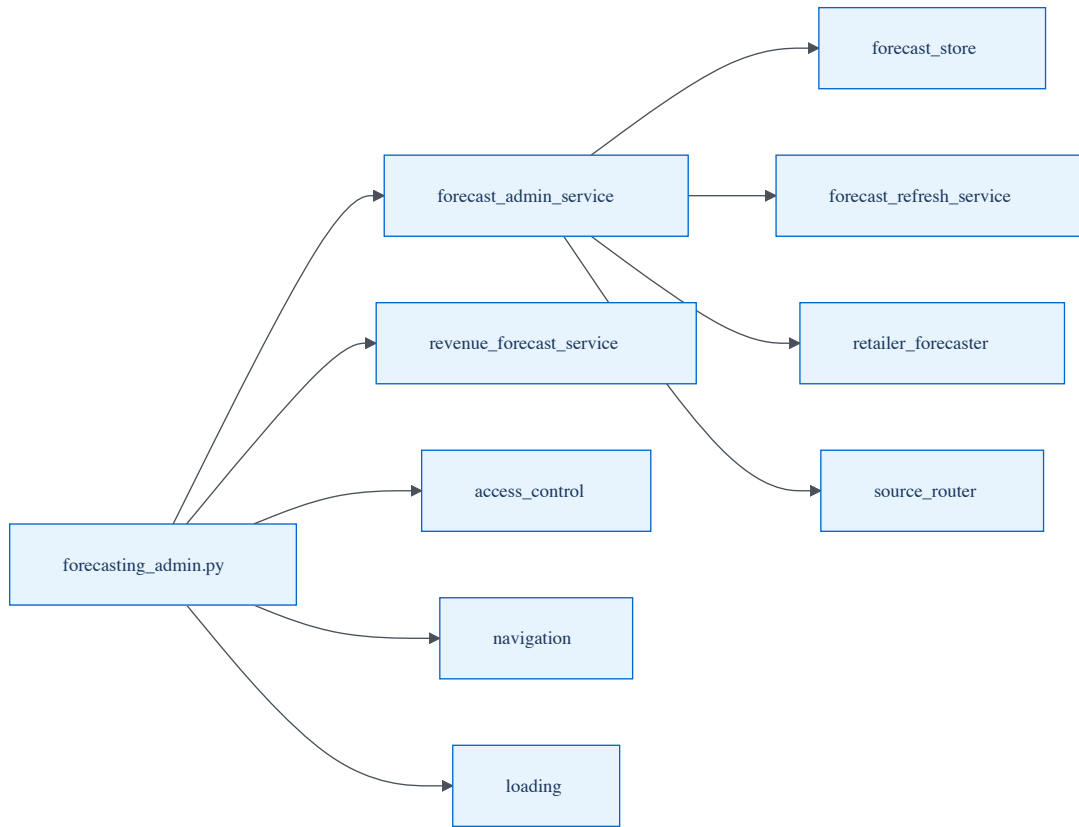
Security Checklist

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File Dependencies

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Generated from source analysis of the Market Data Portal (MDP). Last reviewed against codebase structure as of project documentation pass.